

 RESEARCH & INTERNSHIP REPORT

Public-Key Cryptography & Its Foundations in Number Theory

A mathematical investigation into prime numbers, modular congruences, Fermat & Euler's theorems, and asymmetric encryption algorithms.

Chapter 1: Theoretical Foundations

Connecting ancient mathematical purity with modern digital security & cyber defense architectures.

1.1 Core Terminology of Cryptography

Cryptography (derived from Greek *kryptos* "hidden" and *graphein* "writing") transforms readable data into unintelligible formats.

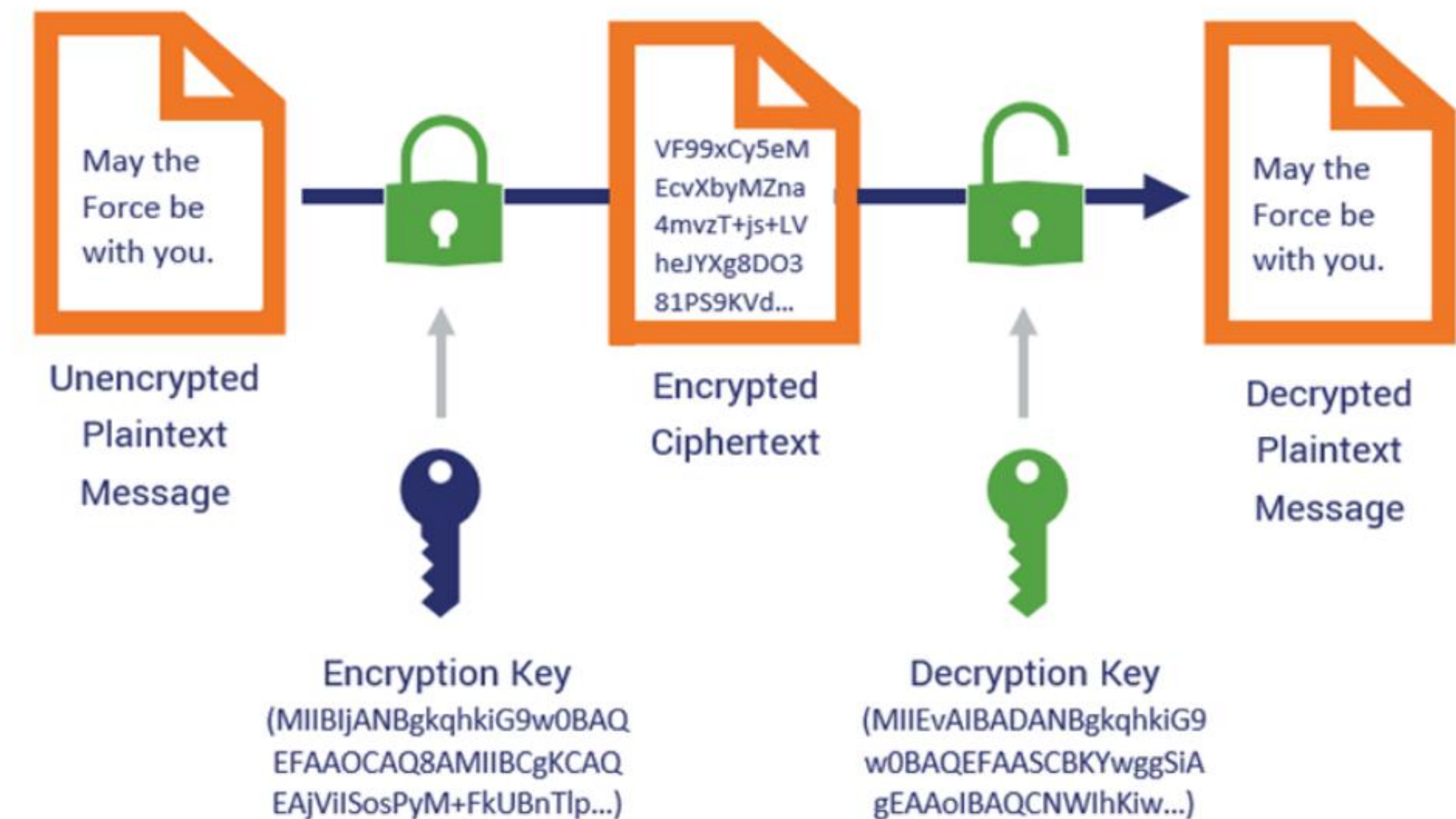
 **Plaintext:** The original readable message given as input to the algorithm.

 **Ciphertext:** The encrypted unreadable output produced by encryption.

 **Key:** A mathematical value used for transforming plaintext to ciphertext & vice versa.

 **Encoder/Decoder:** Electronic devices converting analog and digital signals.

How Asymmetric Encryption Works



1.2 Divisibility & Prime Numbers

Divisibility Property

An integer a divides an integer b (denoted $a \mid b$) if there exists an integer k such that:

$$b = a \times k$$

Divisibility forms the bedrock for congruences, remainders, and prime tests.

Prime Building Blocks

Primes are integers greater than 1 divisible only by 1 and themselves. By the Fundamental Theorem of Arithmetic, every integer decomposes uniquely into primes.

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

Multiplying primes is computationally trivial; factoring large products is computationally intractable.

1.3 GCD & The Euclidean Algorithm



Greatest Common Divisor

The largest integer dividing both a and b is denoted $\gcd(a, b)$. Two integers are **coprime** if $\gcd(a, b) = 1$.



Euclidean Principle

Computes GCD efficiently via division algorithm: $a \bmod b = a - \lfloor a/b \rfloor b$ until remainder reaches zero.



Numerical Calculation

Find :

1. $24 = 18 \times 1 + 6$

2. $18 = 6 \times 3 + 0$

Final non-zero remainder = **6**.

1.4 Modular Arithmetic & Congruence

$$13 \equiv 3$$

mod 5

Since 5 divides $(13 - 3) = 10$ perfectly.

Definition of Congruence

Two integers a and b are congruent modulo m if their difference is a multiple of m :

$$a \equiv b \pmod{m} \Leftrightarrow m \mid (a - b)$$

Arithmetic Operations

- **Addition:** $(a + b) \bmod m = (a \bmod m + b \bmod m) \bmod m$
- **Multiplication:** $(a \cdot b) \bmod m = (a \bmod m \cdot b \bmod m) \bmod m$

1.5 Euler's Totient Function $\phi(n)$

-  **Definition:** $\phi(n)$ counts positive integers up to n that are relatively prime to n .
-  **Prime Number Rule:** For any prime p , every number less than p is coprime to it:
Example: $\phi(7) = 6$ (Coprime set: $\{1, 2, 3, 4, 5, 6\}$).
-  **Composite Calculation:** For $n = 12$, coprime integers are $\{1, 5, 7, 11\}$, so $\phi(12) = 4$.
-  **Multiplicative Product Rule:** If $pq = 1$, then $\phi(pq) = (p - 1)(q - 1)$.
Example: $\phi(35) = \phi(5 \times 7) = (5 - 1)(7 - 1) = 4 \times 6 = 24$.

1.6 Core Theorems in Modular Exponentiation

Theorem	Mathematical Conditions	Formula	Verified Numerical Example
Fermat's Little Theorem	p is prime, $p \nmid a$	$a^{p-1} \equiv 1 \pmod{p}$	Let $a = 2$, $p = 5$: $2^4 = 16 \equiv 1 \pmod{5}$
Fermat's Corollary	p is prime, any integer a	$a^p \equiv a \pmod{p}$	Let $a = 3$, $p = 7$: $3^7 = 2187 \equiv 3 \pmod{7}$
Euler's Generalization	a and n are coprime	$a^{\phi(n)} \equiv 1 \pmod{n}$	Let $a = 3$, $n = 10$: $3^4 = 81 \equiv 1 \pmod{10}$

2.1 Asymmetric Cryptography

Public Key Cryptography utilizes mathematically linked key pairs: a **Public Key** for encryption and a **Private Key** for decryption.

Four Security Pillars

- **Confidentiality:** Only keyholders decrypt data.
- **Integrity:** Detects unauthorized modifications.
- **Authentication:** Verifies user identity.
- **Non-repudiation:** Prevents sender denial.



2.2 Shift Cipher Numerical Examples

Standard Example

Input: NUMBER THEORY IS EASY

Key: $K = 3$

Formula:

$$C = (P + 3) \bmod 26$$

Cipher: QXPEHU WKHRUPLV
HDVB

Caesar Baseline

Input: HELLO

Key: $K = 3$

Shift: $H \rightarrow K, E \rightarrow H, L \rightarrow O$

Cipher: KHOOR

Test Model 1

Input: SITA WAS FOUND IN
LANKA

Key: $K = 2$ (Hanuman $\rightarrow$ Rama)

Shift: $S \rightarrow U, I \rightarrow K, T \rightarrow V$

Cipher: UKVC YCU HQWPF KP
NCPMC

Test Model 2

Input: THOR DEFEATED
THANOS...

Key: $K = 2$

Shift: $T \rightarrow V, H \rightarrow J, O \rightarrow Q$

Cipher: VJQT FGHGCVGF
VJCPQU...

2.3 Diffie-Hellman Key Exchange Computation

Numerical trace for prime $p = 23$ and primitive root generator $g = 5$:

Alice Public Key A

8

$$5^6 \bmod 23 = 15625 \bmod 23$$

Bob Public Key B

19

$$5^{15} \bmod 23$$

Alice Secret Key K

2

$$19^6 \bmod 23 = 47045881 \bmod 23$$

Bob Secret Key K

2

$$8^{15} \bmod 23 = 35184372088832 \bmod 23$$

Both parties compute identical secret key $K = 2$ without transmitting private keys $a = 6$ or $b = 15$.

Questions & Answers

Number theory elevates theoretical mathematics into the backbone of cybersecurity, digital signatures, and secure internet communication.

 [Public-Key Cryptography Presentation](#)

Image Sources



<https://www.thesslstore.com/blog/wp-content/uploads/2020/12/how-asymmetric-encryption-works.png>

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